

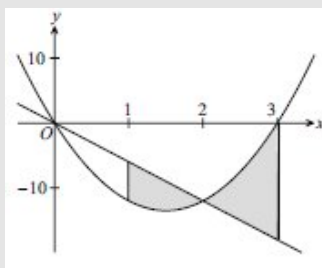
8-4

1. What is the total area of the regions between the curves $y = 6x^2 - 18x$ and $y = -6x$ from $x = 1$ to $x = 3$?

- (A) 4
(B) 12
(C) 16
(D) 20



Answer B



This option is correct. It is helpful to sketch the graphs of the two functions and to check if the two graphs cross inside the given interval. The first graph is a concave up parabola with x -zeros at $x = 0$ and $x = 3$, and the second graph is a line passing through the origin. The sketch indicates that there are two regions between the curves on the interval $1 \leq x \leq 3$. The line intersects the parabola where $6x^2 - 18x = -6x$, or $0 = 6x^2 - 12x = 6x(x - 2)$; therefore the top and the bottom curve of the region switch roles at $x = 2$. Different integrals should be used to find the total area of the region, one for the interval $1 \leq x \leq 2$ and one for the interval $2 \leq x \leq 3$. Let $f(x) = 6x^2 - 18x$ and $g(x) = -6x$; total area is:

$$\begin{aligned} \int_1^3 |f(x) - g(x)| dx &= \int_1^2 |(g(x) - f(x))| dx + \int_2^3 |f(x) - g(x)| dx \\ &= \int_1^2 (12x - 6x^2) dx + \int_2^3 (6x^2 - 12x) dx = (6x^2 - 2x^3) \Big|_1^2 + (2x^3 - 6x^2) \Big|_2^3 \\ &= ((24 - 16) - (6 - 2)) + ((54 - 54) - (16 - 24)) = 4 + 8 = 12. \end{aligned}$$

2. If $0 < c < 1$, what is the area of the region enclosed by the graphs of $y = 0$, $y = 1/x$, $x = c$ and $x = 1$?

- (A) $\ln(1 - c)$
(B) $\ln(1/c)$
(C) $\ln c$
(D) $(1/c^2) - 1$
(E) $1 - (1/c^2)$



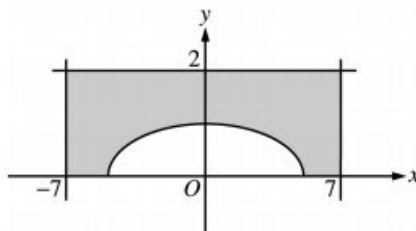
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3.  What is the area of the region enclosed by the graphs of $y = \sqrt{4x - x^2}$ and $y = \frac{x}{2}$?

(A) 1.707
(B) 2.829
(C) 5.389
(D) 8.886
(E) 21.447



4.



The shaded region in the figure above is bounded by the graph of $y = \sqrt{\cos\left(\frac{\pi x}{10}\right)}$ and the lines $x = -7$, $x = 7$, and $y = 2$. What is the area of this region?

(A) 6.372
(B) 7.628
(C) 20.372
(D) 21.634
(E) 24.923

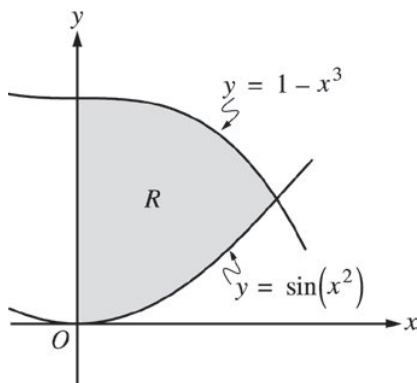


5. What is the area of the region enclosed by the graphs of $f(x) = x - 2x^2$ and $g(x) = -5x$?

(A) $\frac{7}{3}$
(B) $\frac{16}{3}$
(C) $\frac{20}{3}$
(D) 9
(E) 36



8-4



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

6. Find area of R .

Part A

The response can earn up to 2 points:

1 point: For the integrand

$$\text{Area} = \int_0^A (1 - x^3 \sin(x^2)) dx$$

1 point: For the correct answer

$$\text{Area} = 0.533 \text{ (or } 5.34)$$



0	1	2
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The response earns both of the following points:

1 point: For the integrand

$$\text{Area} = \int_0^A (1 - x^3 \sin(x^2)) dx$$

1 point: For the correct answer

$$\text{Area} = 0.533 \text{ (or } 5.34)$$

Limits

Can earn this point only once in the overall scenario:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$

8-4



0

1

The response earns the following point:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$

8-4

7.  A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

Part B

The response can earn up to 3 points:

$$k = B = 0.552352$$

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or 0.256)}$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or 0.276)}$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

Note: Stating value of k does not earn a point.



0	1	2	3
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The response earns all three of the following points:

$$k = B = 0.552352$$

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or 0.256)}$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or 0.276)}$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

Note: Stating value of k does not earn a point.

Limits

8-4

Can earn this point only once in the overall scenario:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$



0	1
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The response earn the following point:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$

8.  What is the area of the region in the first quadrant enclosed by the graphs of $y = \sin(2x)$ and $y = x$?

(A) 0.208

(B) 0.210



(C) 0.266

(D) 0.660

(E) 0.835

9.  What is the area of the region enclosed by the graphs of $y = e^x - 2$, $y = \sin x$, and $x = 0$?

(A) 0.239

(B) 0.506

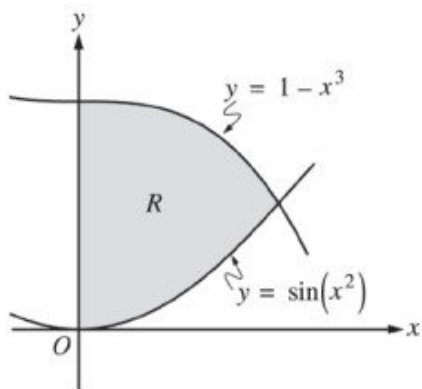
(C) 0.745



(D) 2.340

(E) 3.472

8-4



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

10. Find the area of R .

Part A

One point is earned for the correct limits in an integral in (a), (b), or (c)

One point is earned for the integrand

One point is earned for the answer

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$



0	1	2
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The student earns two of the following points:

One point is earned for the correct limits in an integral in (a), (b), or (c)

One point is earned for the integrand

One point is earned for the answer

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$

8-4

11.  A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

Part B

One point is earned for the integral(s) with k value

One point is earned for the value(s) of integral(s)

One point is earned for the conclusion tied to part (a) or comparison of two integrals

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

The two regions have unequal areas.



0	1	2	3
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The student earns all of the following points:

One point is earned for the integral(s) with k value

One point is earned for the value(s) of integral(s)

One point is earned for the conclusion tied to part (a) or comparison of two integrals

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

The two regions have unequal areas.

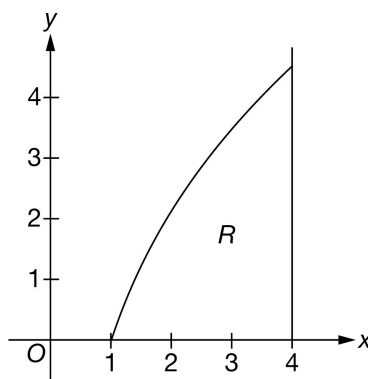
8-4

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region bounded by the graph of $y = \frac{3(x-1)}{\sqrt{x}}$, the x -axis, and the vertical line $x = 4$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.

Part A

The first point is for the integrand only. At most 3 out of 4 points earned for a copy error in the integrand. Subsequent points may be earned.

The second and third points are for antiderivatives. At most 1 out of 2 antiderivative points earned if only one antiderivative is correct OR there is at most one arithmetic sign error in the antiderivatives.

The fourth point is for the limits of integration and evaluation.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ integrand
- ☐ antiderivatives (no criteria)
- ☐ answer

8-4

Solution:

$$\begin{aligned}
 \text{Area} &= \int_1^4 \frac{3(x-1)}{\sqrt{x}} dx = \int_1^4 \left(3\sqrt{x} - \frac{3}{\sqrt{x}} \right) dx \\
 &= \left(3 \cdot \frac{2}{3} x^{\frac{3}{2}} - 3 \cdot 2x^{\frac{1}{2}} \right) \bigg|_{x=1}^{x=4} = \left(2x^{\frac{3}{2}} - 6\sqrt{x} \right) \bigg|_{x=1}^{x=4} \\
 &= (16 - 12) - (2 - 6) = 8
 \end{aligned}$$

Part B

The first point is for the integrand only. At most 4 out of 5 points earned for a copy error in the integrand. Subsequent points may be earned.

The second point is for limits and π .

The third and fourth points are for antiderivatives. At most 1 out of 2 antiderivative points earned if only one antidifferentiation error OR there is at most one arithmetic sign error in the antiderivatives.

The fifth point is for the evaluation, which does not need to be simplified. An error with the factor of 9 applies to this point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4	5
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The student response accurately includes all five of the criteria below.

- ☐ integrand
- ☐ limits and constant
- ☐ antiderivatives (no criteria)
- ☐ answer

Solution:

$$\begin{aligned}
 \text{Volume} &= \pi \int_1^4 \left(\frac{3(x-1)}{\sqrt{x}} \right)^2 dx = 9\pi \int_1^4 \left(\frac{x^2 - 2x + 1}{x} \right) dx \\
 &= 9\pi \int_1^4 \left(x - 2 + \frac{1}{x} \right) dx = 9\pi \left(\frac{x^2}{2} - 2x + \ln x \right) \bigg|_{x=1}^{x=4} \\
 &= 9\pi \left((8 - 8 + \ln 4) - \left(\frac{1}{2} - 2 + 0 \right) \right) = 9\pi \left(\ln 4 + \frac{3}{2} \right)
 \end{aligned}$$

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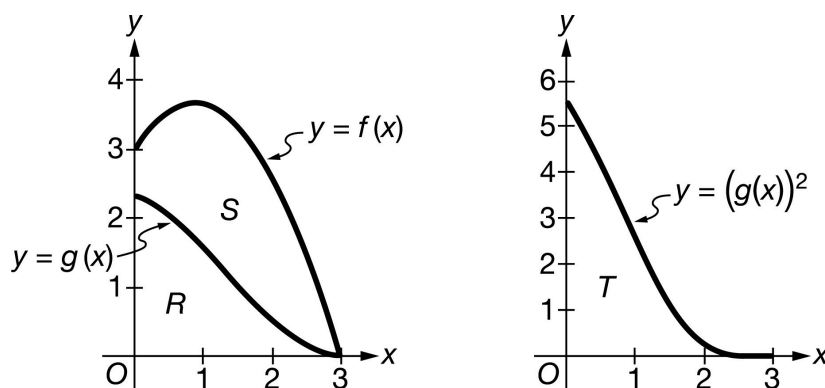
13. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The function f is defined by $f(x) = 3(1+x)^{0.5} \cos\left(\frac{\pi x}{6}\right)$ for $0 \leq x \leq 3$. The function g is continuous and decreasing for $0 \leq x \leq 3$ with $g(3) = 0$.

The figure above on the left shows the graphs of f and g and the regions R and S . R is the region bounded by the graph of g and the x - and y -axes. Region R has area 3.24125. S is the region bounded by the y -axis and the graphs of f and g .

The figure above on the right shows the graph of $y = (g(x))^2$ and the region T . T is the region bounded by the graph of $y = (g(x))^2$ and the x - and y -axes. Region T has area 5.32021.

 (a) Find the area of region S .

(b) Find the volume of the solid generated when region S is revolved about the horizontal line $y = -3$.

(c) Region S is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 7 times the length of its base in region S . Write, but do not evaluate, an integral expression for the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned with either $\int_0^3 (f(x) - g(x)) dx$ or $\int_0^3 f(x) dx$.

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The second point is earned for evidence of 3.24125 being used as an area, which means it is being subtracted outside of an integrand. For example, $\int_0^3 f(x) dx - 3.24125$ would earn the point, while $\int_0^3 f(x) - 3.24125$ would provide evidence for earning this point only in the presence of the final answer of 4.920 (or 4.919).

Note that responses in degree mode, which lead to a final answer of approximately 10.756727, do not earn the first point for which they are eligible. Thus a fully correct solution in degree mode would earn two of the three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ definite integral of f
- ☐ uses area of R
- ☐ answer

Solution:

$$\begin{aligned}\int_0^3 (f(x) - g(x)) dx &= \int_0^3 f(x) dx - \int_0^3 g(x) dx \\ &= \int_0^3 f(x) dx - 3.24125 = 4.919585\end{aligned}$$

The area of region S is 4.920 (or 4.919).

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for an integrand of the form $(f(x) - k)^2 - (g(x) - k)^2$ where $k = -3$, 0, or 3.

The second point can be earned if f and g are reversed.

The third point is earned for replacing $\int_0^3 g(x) dx$ with 3.24135 and $\int_0^3 (g(x))^2 dx$ with 5.32021. There are multiple ways that a response can use these values correctly, so the third point can also be earned with sufficient work in the presence of the correct final value of 156.263709.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ form of integrand
- ☐ integrand
- uses areas of R and T

8-4

☐☐

limits, constant, and answer

Solution:

$$\begin{aligned}
 & \pi \int_0^3 \left((f(x) + 3)^2 - (g(x) + 3)^2 \right) dx \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - \int_0^3 ((g(x))^2 + 6g(x) + 9) dx \right) \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - \int_0^3 (g(x))^2 dx - 6 \int_0^3 g(x) dx - \int_0^3 9 dx \right) \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - 5.32021 - 6 \cdot 3.24125 - 9 \cdot 3 \right) \\
 &= 156.263709
 \end{aligned}$$

The volume of the solid is 156.264 (or 156.263).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an integrand of the form $(f(x) - g(x))^2$.

0	1	2
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The student response accurately includes **both** of the criteria below.☐

integrand

☐

expression

Solution:

$$\text{Volume} = \int_0^3 7(f(x) - g(x))^2 dx$$

8-4

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

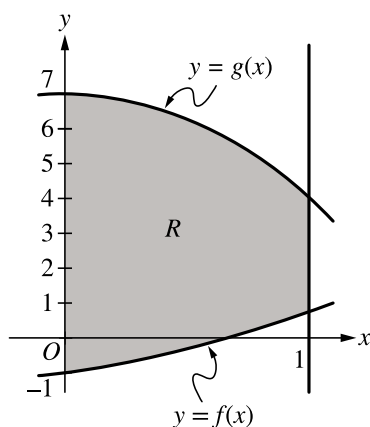


Figure 1

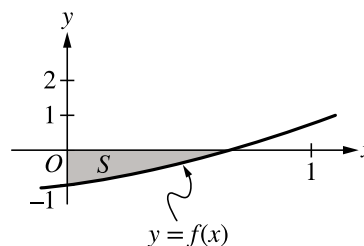


Figure 2

Let f and g be the functions given by $f(x) = e^x - 2$ and $g(x) = -3x^2 + 7$. Let R be the shaded region bounded by the graphs of f and g , the y -axis, and the vertical line $x = 1$, as shown in Figure 1. Let S be the shaded region in Quadrant IV bounded by the graph of f , the x -axis, and the y -axis, as shown in Figure 2.

- Find the area of R .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 8$.
- The region S is the base of a solid. For this solid, each cross section perpendicular to the y -axis is a rectangle with height 5 and base in region S . Write, but do not evaluate, an integral expression that gives the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $g(x) - f(x)$, $f(x) - g(x)$, $|g(x) - f(x)|$, or $|f(x) - g(x)|$ regardless of any subsequent copy errors in presentation.

- A response that presents any subsequent copy or parentheses errors is eligible for the second and third points, but not the fourth.

A response earns the fourth point only if it has earned the first 3 points and presents an answer equivalent to $9 - e$.

A response that uses an integrand of $f(x) - g(x)$:

- can earn the fourth point if it presents correct communication, such as $\int_0^1 (f(x) - g(x)) dx = e - 9$, so the area is $9 - e$.

8-4

- does not earn the fourth point if it includes incorrect communication, such as $\int_0^1 (f(x) - g(x)) \, dx = 9 - e$.

Special Case: A response that does not earn the first point, but presents $\int_0^1 (-3x^2 + 7 - e^x - 2) \, dx$ (or equivalent), is eligible for the second and third points, and earns the fourth point only for an answer of $5 - e$.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative of $3x^2$ or e^x
- ☐ Remaining antiderivatives
- ☐ Answer

Solution:

$$\begin{aligned}
 \text{Area} &= \int_0^1 (g(x) - f(x)) \, dx \\
 &= \int_0^1 ((-3x^2 + 7) - (e^x - 2)) \, dx = \int_0^1 (-3x^2 + 9 - e^x) \, dx \\
 &= [-x^3 + 9x - e^x]_0^1 = (-1 + 9 - e) - (-1) = 9 - e
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting both the correct constant and correct limits of integration in an integral expression involving both $f(x)$ and $g(x)$.

The second point is earned by an integrand of the form $R^2 - r^2$, where one of R and r is of the form $(\pm a \pm f(x))$ and the other is of the form $(\pm a \pm g(x))$, for $a \neq 0$.

The third point is earned only for the correct integrand.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits and constant
- ☐ Form of integrand
- ☐ Integrand

8-4

Solution:

$$\text{Volume} = \pi \int_0^1 \left((8 - f(x))^2 - (8 - g(x))^2 \right) dx$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Alternate solution:

$$f(x) = 0 \Rightarrow x = \ln 2$$

$$\text{Volume} = 5 \cdot \text{Area}(S) = 5 \int_0^{\ln 2} (-f(x)) dx$$

These responses earn 1 of the 2 points:

- $5 \int_{-1}^0 \ln(y+2) dx$
- $5 \int_0^{\ln 2} (-f(x)) dy,$
- $5 \int_0^{\ln 2} -(e^x - 2) dy$
- $5 \int_{-1}^0 \ln(y+2)$
- $5 \int_0^{\ln 2} -f(x), 5 \int_0^{\ln 2} -(e^x - 2)$
- $5 \int_{-1}^0 -f(x) dy$
- $5 \int_0^{\ln 2} \ln(y+2) dx$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Limits and constant
- ☐ Integrand

Solution:

$$y = f(x) = e^x - 2 \Rightarrow x = \ln(y + 2)$$

8-4

$$\text{Volume} = \int_{-1}^0 5 \ln(y + 2) \, dy$$

8-4

15. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let R be the region in the first quadrant enclosed by the graphs of $y = e^{(x^2)} - 1$ and $y = 4x$.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The first point requires a definite integral with a correct integrand. At most 2 out of 3 points earned for a copy error in the integrand **OR** a reversal of terms in the difference in the integrand. Subsequent points may be earned.

The second point requires correct antidifferentiation.

The third point is for limits and evaluation.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

Solution:

8-4

$$\text{Area} = \int_0^4 (4x - x^2) dx = \left(2x^2 - \frac{x^3}{3} \right) \bigg|_{x=0}^{x=4} = 32 - \frac{64}{3} = \frac{32}{3}$$

Part B

The first point requires a difference of squares for the integrand.

The second point requires both terms of the integrand.

At most 1 out of 2 integrand points earned if there is one error in the integrand OR a reversal of terms.

The third point requires the limits and π .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integrand of difference of squares
- ☐ integrand terms
- ☐ limits and π

Solution:

$$\text{Volume} = \pi \int_0^4 \left((16 - x^2)^2 - (16 - 4x)^2 \right) dx$$

Part C

The first point requires an integrand in terms of y that addresses the length of a side of the square cross section.

The second point requires the integrand to represent the cross sectional area.

At most 1 out of 2 integrand points earned if there is one error in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integrand in terms of y for square cross section
- ☐ integrand terms
- ☐ limits

8-4**Solution:**

$$\text{Volume} = \int_0^{16} \left(\sqrt{y} - \frac{y}{4} \right)^2 dy$$

8-4

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let R be the region in the first quadrant bounded by the graph of $y = \frac{1}{x}$, the horizontal line $y = 1$, and the vertical line $x = e$.

- Find the area of region R .
- Find the volume of the solid generated when region R is revolved about the line $y = 1$.
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a semicircle. Write, but do not evaluate, an expression involving an integral that gives the volume of the solid.

Part A

At most 2 out of 3 points earned if there is one error in the integrand **OR** 1 error with limits of integration. Numerical answers do not need to be simplified (e.g., $\ln e$)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ definite integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\text{Area} = \int_1^e \left(1 - \frac{1}{x}\right) dx = (x - \ln x) \Big|_1^e = (e - \ln e) - (1 - \ln 1) = e - 2$$

Part B

At most 2 out of 3 points earned if there is one error in the integrand **OR** 1 error with limits of integration. Numerical answers do not need to be simplified (e.g., $\ln e$)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-4



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand
- ☐ limits and π
- ☐ antiderivative
- ☐ answer

Solution:

$$\begin{aligned}\text{Volume} &= \pi \int_1^e \left(1 - \frac{1}{x}\right)^2 dx = \pi \int_1^e \left(1 - \frac{2}{x} + \frac{1}{x^2}\right) dx \\ &= \pi \left(x - 2 \ln x - \frac{1}{x}\right) \Big|_1^e = \pi \left(e - 2 \ln e - \frac{1}{e}\right) = \pi \left(e - 2 - \frac{1}{e}\right)\end{aligned}$$

Part C

Halving the diameter of the semicircle to find an expression for its radius may appear in the integrand, as presented. It may also appear as part of the constant multiple.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrand
- ☐ limits and constant

Solution:

$$\text{Volume} = \frac{\pi}{2} \int_1^e \left(\frac{1}{2} \left(1 - \frac{1}{x}\right)\right)^2 dx$$

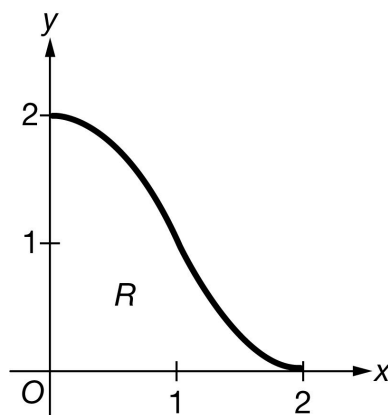
8-4

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be the function defined by $f(x) = \begin{cases} 2 - x^2 & \text{for } 0 \leq x < 1 \\ (x - 2)^2 & \text{for } 1 \leq x \leq 2 \end{cases}$. Let R be the region in the first quadrant bounded by the graph of f and the x - and y -axes, as shown in the figure above.

- Find the area of R .
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.
- The portion of the region R for $1 \leq x \leq 2$ is revolved about the y -axis to form a solid. Find the volume of the solid.

Part A

At most 2 out of 3 points earned for a copy error in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ definite integrals
- ☐ antiderivatives
- ☐ answer

8-4

Solution:

$$\begin{aligned}
 \text{Area} &= \int_0^1 (2 - x^2) \, dx + \int_1^2 (x - 2)^2 \, dx \\
 &= \left(2x - \frac{1}{3}x^3\right) \Big|_0^1 + \frac{1}{3}(x - 2)^3 \Big|_1^2 \\
 &= \left(\frac{5}{3} - 0\right) + \left(0 - \left(-\frac{1}{3}\right)\right) = 2
 \end{aligned}$$

Part B

At most 1 out of 2 points earned if there is one error in the integrand OR 1 error with limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrands
- ☐ expression for total volume

Solution:

$$\text{Volume} = \int_0^1 (2 - x^2)^2 \, dx + \int_1^2 (x - 2)^4 \, dx$$

Part C

Third and fourth points may be earned if there is a maximum of one error in either the integrand OR one error in the limits and constant. Eligibility requires a difference of squares for the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand
- ☐ limits and π
- ☐ antiderivative
- ☐ answer

Solution:

8-4

$$y = (x - 2)^2 \Rightarrow x = \sqrt{y} + 2$$

$$\begin{aligned}\text{Volume} &= \pi \int_0^1 \left((\sqrt{y} + 2)^2 - 1^2 \right) dy = \pi \int_0^1 (y + 4\sqrt{y} + 4 - 1) dy \\ &= \pi \int_0^1 (y + 4\sqrt{y} + 3) dy = \pi \left(\frac{y^2}{2} + \frac{8}{3} y^{\frac{3}{2}} + 3y \right) \Big|_0^1 \\ &= \pi \left(\frac{1}{2} + \frac{8}{3} + 3 \right) = \frac{37\pi}{6}\end{aligned}$$

8-4

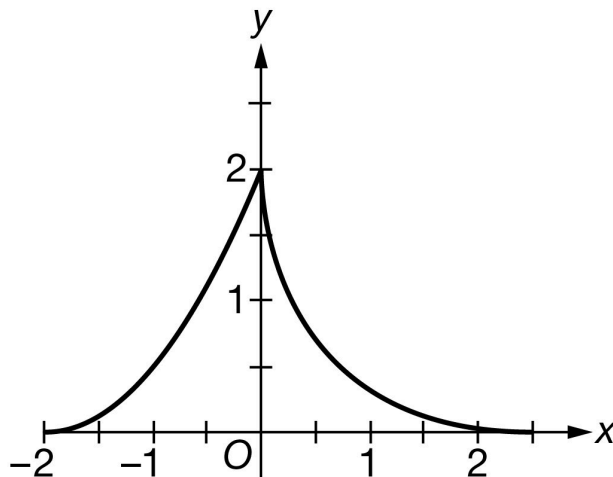
18.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be the function defined by $f(x) = \begin{cases} \frac{1}{2}(x+2)^2 & \text{for } -2 \leq x < 0 \\ 2 - 2\sin\sqrt{x} & \text{for } 0 \leq x \leq \frac{\pi^2}{4} \end{cases}$. The graph of f is shown in the figure above. Let R be the region bounded by the graph of f and the x -axis.

- Find the area of R .
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.
- The portion of the region R for $1 \leq y \leq 2$ is revolved about the x -axis to form a solid. Find the volume of the solid.

Part A

There are closed form expressions for the antiderivatives. The response may give an exact value for the area, which is $\frac{4}{3} + \frac{1}{2}(\pi^2 - 8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-4



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ definite integral for $-2 \leq x < 0$
- ☐ definite integral for $0 \leq x \leq \frac{\pi^2}{4}$
- ☐ answer

Solution:

$$\text{Area} = \int_{-2}^0 \frac{1}{2}(x+2)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2 \sin \sqrt{x}) dx = 2.268$$

Part B

At most 1 out of 2 points earned if there is one error in the integrand OR 1 error with limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrands
- ☐ expression for total volume

Solution:

$$\text{Volume} = \int_{-2}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2 \sin \sqrt{x})^2 dx$$

Part C

To be eligible for the fourth point, the response must have earned the first two points.

Special case: A response with the expression

$$\pi \int_{-0.585786}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \pi \int_0^{0.274156} (2 - 2 \sin \sqrt{x})^2 dx = 5.688 \text{ (or } 5.687)$$

earns the third and fourth points.

8-4

There are closed form expressions for the antiderivatives. The response may give an exact values for the limits and volume.

The limits are $x_1 = \sqrt{2} - 2$ and $x_2 = \frac{\pi^2}{36}$. An unsimplified expression for the volume is given by

$$\frac{\pi}{20} \left(2^5 - (\sqrt{2})^5 \right) + \pi (\sqrt{2} - 2) + \frac{\pi}{36} (-270 + 42\sqrt{3}\pi + 5\pi^2).$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand for one region
- ☐ integrand for the remaining region
- ☐ limits
- ☐ constant and answer

Solution:

$$1 = \frac{1}{2}(x_1 + 2)^2 \Rightarrow x_1 = -0.585786$$

$$1 = 2 - 2 \sin \sqrt{x_2} \Rightarrow x_2 = 0.274156$$

$$\begin{aligned} \text{Volume} &= \pi \int_{x_1}^0 \left(\left(\frac{1}{2}(x+2)^2 \right)^2 - 1^2 \right) dx + \pi \int_0^{x_2} \left((2 - 2 \sin \sqrt{x})^2 - 1^2 \right) dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

8-4

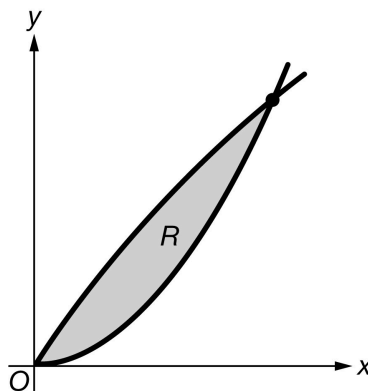
19.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region in the first quadrant enclosed by the graphs of $y = \arctan x$ and $y = x^2$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is a rectangle whose height is twice the length of its base in region R . Find the volume of the solid.

Part A

The first point requires a definite integral with a correct integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrand
limits and answer

8-4

**Solution:**

The graphs of $y = \arctan x$ and $y = x^2$ intersect at $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (0.833606, 0.694899)$.

$$\text{Area} = \int_{x_1}^{x_2} (\arctan x - x^2) dx = 0.122$$

Part B

The first point requires a definite integral with a difference of squares for the integrand.

The second point requires both terms of the integrand.

At most 1 out of 2 integrand points earned if there is one error in the integrand OR a reversal of terms.

A response that imports incorrect limits from part (a) is eligible for all points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integrand of difference of squares
- ☐ integrand terms
- ☐ limits, constant, and answer

Solution:

From part (a), $x_1 = 0$ and $x_2 = 0.833606$.

$$\text{Volume} = \pi \int_{x_1}^{x_2} ((\arctan x)^2 - (x^2)^2) dx = 0.073\pi = 0.230 \text{ (or } 0.229)$$

Part C

The first point requires an integrand in terms of y that addresses the length of either side of the rectangular cross section.

The second point requires the integrand to represent the cross sectional area.

At most 1 out of 2 integrand points earned if there is one error in the integrand.

A response is eligible for all points if the limits are consistent with the x -values from part (a).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-4



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand in terms of y for rectangular cross section
- ☐ integrand terms
- ☐ limits
- ☐ answer

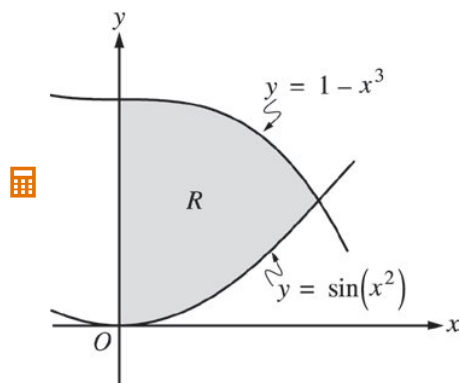
Solution:

From part (a), $y_1 = 0$ and $y_2 = 0.694899$.

$$\text{Volume} = \int_{y_1}^{y_2} 2(\sqrt{y} - \tan y)^2 dy = 0.050 \text{ (or } 0.049)$$

8-4

20.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

(a) Find the area of R .

(b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

(c) Find the volume of the solid generated when R is revolved about the line $y = -3$.

Part A

1 point: For the integrand

1 point: For the correct answer

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integrand

1 point: For the correct answer

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

8-4

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

Part B

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

Note: Stating value of k does not earn a point.

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

Note: Stating value of k does not earn a point.

8-4

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

Part C

2 points: For the correct integrand

1 point: For the correct answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841 \text{ (or } 11.840)\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

2 points: For the correct integrand

1 point: For the correct answer

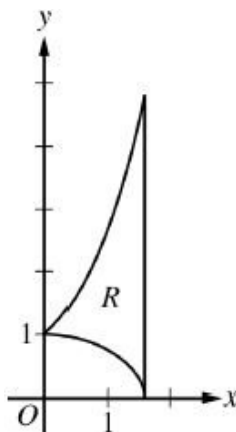
$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841 \text{ (or } 11.840)\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

8-4

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Let R be the region in the first quadrant enclosed by the graph of $f(x) = \sqrt{\cos x}$, the graph of $g(x) = e^x$, and the vertical line $x = \frac{\pi}{2}$, as shown in the figure above.

- Write, but do not evaluate, an integral expression that gives the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R is the base of a solid whose cross sections perpendicular to the x -axis are semicircles with diameters on the xy -plane. Write, but do not evaluate, an integral expression that gives the volume of this solid.

Part A**1 point is earned for:** integrand**1 point is earned for:** limits

$$\text{Area} = \int_0^{\pi/2} (g(x) - f(x)) dx$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integrand**1 point is earned for:** limits

$$\text{Area} = \int_0^{\pi/2} (g(x) - f(x)) dx$$

Part B**1 point is earned for:** integrand**2 points are earned for:** antiderivatives**1 point is earned for:** answer

8-4

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\pi/2} \left((g(x))^2 - (f(x))^2 \right) dx \\
 &= \pi \int_0^{\pi/2} \left((e^x)^2 - (\sqrt{\cos x})^2 \right) dx \\
 &= \pi \int_0^{\pi/2} (e^{2x} - \cos x) dx = \pi \left[\frac{1}{2} e^{2x} - \sin x \right]_{x=0}^{x=\pi/2} \\
 &= \pi \left(\frac{1}{2} e^{\pi} - \sin \frac{\pi}{2} - \left(\frac{1}{2} - 0 \right) \right) = \pi \left(\frac{1}{2} e^{\pi} - \frac{3}{2} \right)
 \end{aligned}$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: integrand

2 points are earned for: antiderivatives

1 point is earned for: answer

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\pi/2} \left((g(x))^2 - (f(x))^2 \right) dx \\
 &= \pi \int_0^{\pi/2} \left((e^x)^2 - (\sqrt{\cos x})^2 \right) dx \\
 &= \pi \int_0^{\pi/2} (e^{2x} - \cos x) dx = \pi \left[\frac{1}{2} e^{2x} - \sin x \right]_{x=0}^{x=\pi/2} \\
 &= \pi \left(\frac{1}{2} e^{\pi} - \sin \frac{\pi}{2} - \left(\frac{1}{2} - 0 \right) \right) = \pi \left(\frac{1}{2} e^{\pi} - \frac{3}{2} \right)
 \end{aligned}$$

Part C

1 point is earned for: integrand

1 point is earned for: limits

1 point is earned for: constant

$$\text{Volume} = \frac{1}{2} \int_0^{\pi/2} \pi \left(\frac{g(x) - f(x)}{2} \right)^2 dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integrand

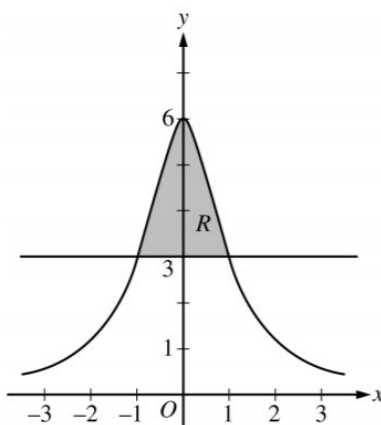
1 point is earned for: limits

8-4**1 point is earned for:** constant

$$\text{Volume} = \frac{1}{2} \int_0^{\pi/2} \pi \left(\frac{g(x) - f(x)}{2} \right)^2 dx$$

8-4

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Let f be the function defined by $f(x) = \frac{6}{1+x^2}$. Let R be the shaded region bounded by the graph of f and the horizontal line $y = 3$, as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

(c) Let $h(x)$ be the vertical distance between the point $(x, f(x))$ and the horizontal line $y = 3$. Find the rate of change of $h(x)$ with respect to x at $x = 2$.

Part A

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral

8-4

1 point is earned for: antiderivative**1 point is earned for:** answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.**Part B****2 points are earned for:** integrand**1 point is earned for:** limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for: integrand**1 point is earned for:** limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) dx$$

Part C**1 point is earned for:** $h(x)$ **1 point is earned for:** $h'(x)$ **1 point is earned for:** $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

8-4

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $h(x)$

1 point is earned for: $h'(x)$

1 point is earned for: $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$

8-4

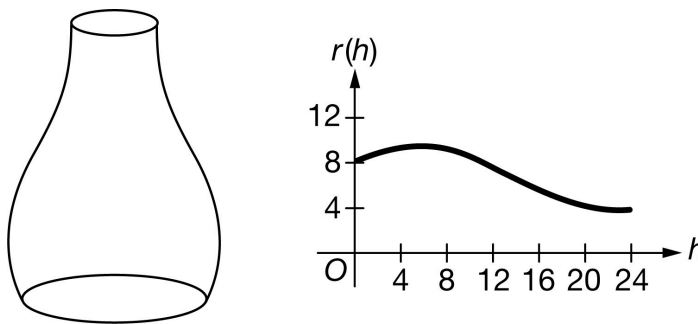
23.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



A 24-centimeter tall vase has cross sections parallel to the base that are circles. Each circular cross section h centimeters above the base has a radius of $r(h) = 8 - 0.1h + 2 \sin(0.3h^{0.9})$ centimeters for $0 \leq h \leq 24$. A sketch of the vase and the graph of r are shown above.

- (a) Find the area of the region between the graph of r and the h -axis from $h = 0$ to $h = 24$.
- (b) Find the volume of the vase.

(c) Evaluate $\int_4^{20} r'(h) \, dh$. Using correct units, interpret the meaning of $\int_4^{20} r'(h) \, dh$ in the context of the problem.

(d) Water is poured into the vase. At the instant when the depth of the water in the vase is 10 centimeters, the depth of the water is increasing at a rate of 0.6 centimeter per second. At that instant, what is the rate of change of the circumference of the surface of the water, in centimeters per second?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An unsupported answer of 165.781 (or 165.78) will not earn any points.

The first point is earned by using $r(h)$ as the integrand in a definite integral. A definite integral with incorrect limits or multiplied by any constant other than 1 will not earn the second point.

Do not read units in this problem.

8-4

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, area = 165.509 (or 165.508).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\int_0^{24} r(h) dh = 165.780506$$

The area of the region is 165.781 (or 165.78).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for writing a definite integral (with any limits) with integrand $c(r)^2$, for any nonzero constant c .

Once the first point is earned, it cannot be lost. Any subsequent copy errors in the presence of the correct answer will be considered a calculator restart and will earn the second point.

To earn the second point, a response must earn the first point, show limits from 0 to 24, multiply by π , and present the correct evaluation.

Special case: A response that fails to earn the first point because of a copy error will earn the second point for the correct answer.

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the second point. (For example, 1252.387π is not correct to three digits after the decimal and therefore does not earn the second point.)

Degree mode: volume = 3616.746 (or 3616.745) (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

8-4

Solution:

$$\int_0^{24} \pi(r(h))^2 dh = \pi \cdot 1252.387790 = 3934.492281$$

The volume of the vase is 3934.492 centimeters.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the value -5.259 ; no supporting work is required.

The interpretation must clearly indicate that the radius at $h = 20$ is smaller than the radius at $h = 4$. Therefore, “the difference in the radius when $h = 20$ and $h = 4 \dots$ ” is not sufficient.

The response, “The difference (or change) in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns both points. However, “The *total* change in radius from $h = 4$ to $h = 20$ is -5.259 cm” earns only the first point.

Degree mode: answer = -1.481 (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation with units

Solution:

$$\int_4^{20} r'(h) dh = r(20) - r(4) = -5.259388$$

The radius of the vase at $h = 20$ centimeters is 5.259 centimeters less than the radius of the vase at $h = 4$ centimeters.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by using $\frac{dr}{dh} \cdot \frac{dh}{dt} \frac{dr}{dt}$ in an expression for $\frac{dC}{dt}$.

A response may work from $C = 2\pi(8 - 0.1h + 2\sin(0.3h^{0.9}))$. In this case, the first point is earned for $\frac{dC}{dt} = \frac{dC}{dh} \cdot \frac{dh}{dt}$.

The second point is earned for using either the value 0.6 as $\frac{dh}{dt}$ or the value -0.411320 as $\frac{dr}{dh}$ in an expression for $\frac{dC}{dt}$. This point is not earned if either 0.6 or $r'(10)$ is used as $\frac{dr}{dt}$.

8-4

The second point can be earned without the first point. For example, a computation or communication error in attempting to use the chain rule for $\frac{dC}{dt}$ does not earn the first point but is eligible for the second point.

A response that uses an incorrect equation for the circumference in terms of r only, such as $C = \pi r^2$, is eligible for the first two points but not the third point.

The third point is earned only for an answer of -1.551 (or -1.55).

A final answer written in terms of π must be correct to three places after the decimal point, rounded or truncated, when converted to a decimal value to earn the third point.

Degree mode: answer = -0.349 (or -0.348) (See degree mode statement in part (a).)



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\frac{dr}{dt} = \frac{dr}{dh} \cdot \frac{dh}{dt}$
 Uses $\frac{dh}{dt} \Big|_{h=10} = 0.6$
☐ – OR –
 Uses $\frac{dr}{dh} \Big|_{h=10} = r'(10)$
☐ Answer

Solution:

$$C = 2\pi r \Rightarrow \frac{dC}{dt} = \frac{d}{dt}(2\pi r) = 2\pi \cdot \left(\frac{dr}{dh} \cdot \frac{dh}{dt}\right)$$

$$\begin{aligned} \frac{dC}{dt} \Big|_{h=10} &= 2\pi \cdot \left(\frac{dr}{dh} \Big|_{h=10} \cdot \frac{dh}{dt} \Big|_{h=10}\right) \\ &= 2\pi \cdot (r'(10) \cdot 0.6) = \pi \cdot (-0.493584) = -1.550641 \end{aligned}$$

The rate of change of the circumference of the surface of the water is -1.551 (or -1.55) centimeters per second.

24. What is the area of the region enclosed by the graphs of $y = \frac{1}{1+x^2}$ and $y = x^2 - \frac{1}{3}$?

(A) 0.786

(B) 0.791

(C) 1.582

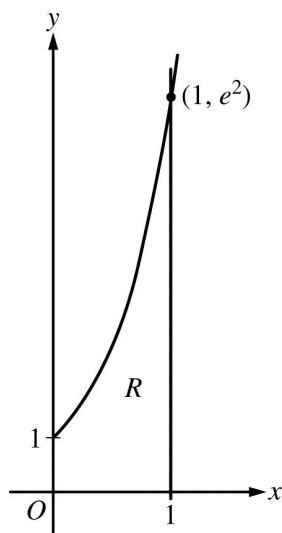
(D) 1.837

(E) 1.862



8-4

25.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

Find the area of R .

Part B

The response can earn up to 3 points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = 1 + \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For the integral

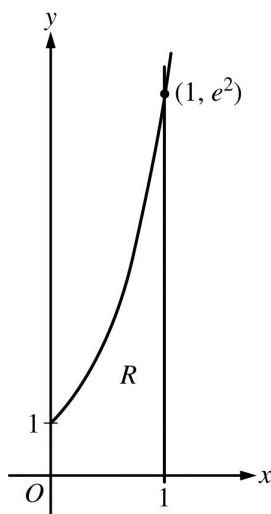
1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = 1 + \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$

8-4

26.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

- Write an equation for the line tangent to the graph of f at $x = 1$.
- Find the area of R .
- Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The response can earn up to 2 points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

8-4

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.

Part B

The response can earn up to 3 points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$$

Part C

The response can earn up to 4 points:

2 points: For the integrand

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y \right)^2 dy$$

8-4



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

2 points: For the integrand

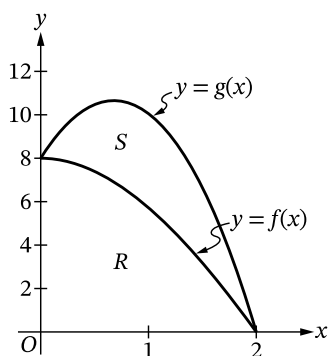
1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

8-4

27. The function f is defined by $f(x) = 8 \cos\left(\frac{\pi}{4}x\right)$, and g is a continuous function such that $\int_0^2 g(x)dx = 16$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ for $0 \leq x \leq 2$ and the x - and y -axes. Let S be the region in the first quadrant bounded by the graphs of $y = f(x)$ and $y = g(x)$ for $0 \leq x \leq 2$. Regions R and S are shown in the figure.

**Part A**

Find the area of region R . Show the work that leads to your answer.

Part B

Find the area of region S .

Part C

Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a triangle with its base in region R and with height equal to the length of its base. Write, but do not evaluate, an integral expression for the volume of the solid.

Part D

For $y = f(x)$ on the interval $0 \leq x \leq 2$, it can be shown that $x = \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)$ for $0 \leq y \leq 8$. Write, but do not evaluate, an integral expression that gives the volume of the solid when R is rotated about the vertical line $x = 3$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a definite integral with an integrand of either $f(x)$ or $8 \cos\left(\frac{\pi}{4}x\right)$, with or without the differential dx . A definite integral with incorrect limits of integration is eligible to earn **P1**.

Model Solution:

$$\text{Area of } R = \int_0^2 f(x) dx$$

8-4



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A definite integral with incorrect limits of integration is eligible to earn **P2**.
- **P2** is earned for an antiderivative of the form $k \sin\left(\frac{\pi}{4}x\right)$ or equivalent, where k is any positive constant.

Model Solution:

$$\int_0^2 8 \cos\left(\frac{\pi}{4}x\right) dx = \frac{32}{\pi} \sin\left(\frac{\pi}{4}x\right) \Big|_0^2$$



0	1
---	---

The response accurately includes the criteria below.

- Form of antiderivative

Part A Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response earns **P3** if it has earned **P1** and **P2**, presents a definite integral with correct limits, and reports an answer equivalent to $\frac{32}{\pi}$.

Model Solution:

The area of region R is $\frac{32}{\pi} \left(\sin\left(\frac{\pi}{2}\right) - \sin 0 \right) = \frac{32}{\pi}$.



0	1
---	---

The response accurately includes the criteria below.

8-4

- Area of R

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for an answer of $16 - [\text{answer from part A}]$, with no supporting work.

Model Solution:

$$\text{Area of } S = 16 - \frac{32}{\pi}$$



0	1
---	---

The response accurately includes the criteria below.

- Area of S

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P5** is earned only for a response with an integrand of the form $k(f(x))^2$ in a definite or indefinite integral, where k is any nonzero constant.

Model Solution:

$$\begin{aligned} \text{Volume} &= \int_0^2 \frac{1}{2}(f(x))^2 dx \\ &= \int_0^2 \frac{1}{2} \left(8 \cos \left(\frac{\pi}{4} x \right) \right)^2 dx = 32 \int_0^2 \cos^2 \left(\frac{\pi}{4} x \right) dx \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Form of integrand

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

8-4

Scoring Notes:

- **P6** is earned for the correct definite integral, with or without the differential dx .
- Any of $\int_0^2 \frac{1}{2}(f(x))^2 dx$, $\int_0^2 \frac{1}{2}\left(8 \cos\left(\frac{\pi}{4}x\right)\right)^2 dx$, $32 \int_0^2 \cos^2\left(\frac{\pi}{4}x\right) dx$, or a mathematically equivalent expression is sufficient to earn **P6**.

Model Solution:

$$\begin{aligned}\text{Volume} &= \int_0^2 \frac{1}{2}(f(x))^2 dx \\ &= \int_0^2 \frac{1}{2}\left(8 \cos\left(\frac{\pi}{4}x\right)\right)^2 dx = 32 \int_0^2 \cos^2\left(\frac{\pi}{4}x\right) dx\end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for a response that presents an integrand of $3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2$, $3^2 - \left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2$, $\left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2$, $\left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2 - 3^2$, or any mathematically equivalent expression, in one or more definite or indefinite integrals.

Model Solution:

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2\right) dy$$



0	1
---	---

The response accurately includes the criteria below.

- Form of integrand

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

8-4

- **P8** is earned only for an integrand mathematically equivalent to $3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2$ or $3^2 - \left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2$ in one or more definite or indefinite integral.
 - For example: $\int_0^8 \left| \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2 \right| dy$ or $\int_0^8 \left(\left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2 \right) dy$ earns **P8**.
 - Note: An integrand of $\left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2$ or $\left(\frac{4}{\pi} \arccos\left(\frac{y}{8}\right) - 3\right)^2 - 3^2$ does not earn **P8** unless the reversal is resolved.

Model Solution:

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right) \right)^2 \right) dy$$



0	1
---	---

The response accurately includes the criteria below.

- Integrand

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible for **P9**, a response must have earned **P7** or **P8** or must have presented an integrand involving $\frac{4}{\pi} \arccos\left(\frac{y}{8}\right)$ of the form $R^2 - r^2$ in a definite integral.
- **P9** is earned only by an eligible definite integral with an integrand in terms of y , including the constant π , limits $y = 0$ to $y = 8$, and differential dy . A response that presents any other constant, limits (including $y = 8$ to $y = 0$, except in the note below), or differential does not earn **P9**.
 - Note: $\pi \int_8^0 \left(\left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right)\right)^2 - 3^2 \right) dy$ earns **P9**.
- The integral may be split into two integrals; for example, $\pi \int_0^8 3^2 dy - \pi \int_0^8 \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right) \right)^2 dy$ or $\pi \cdot 3^2 \cdot 8 - \pi \int_0^8 \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right) \right)^2 dy$.

Model Solution:

$$\text{Volume} = \pi \int_0^8 \left(3^2 - \left(3 - \frac{4}{\pi} \arccos\left(\frac{y}{8}\right) \right)^2 \right) dy$$



0	1
---	---

The response accurately includes the criteria below.

8-4

- Limits, constant, and differential

28. What is the area of the region in the first quadrant bounded by the lines $x = 1$ and $x = 3$ and the graphs of $y = e^x$ and $y = x - 1$?

(A) $e^3 - e - 6$

(B) $e^3 - e - 2$ ✓

(C) $e^3 - 3$

(D) $e^3 - e$

Answer B

Correct. The graph of $y = e^x$ lies above the graph of $y = x - 1$ over the region in the first quadrant between the lines $x = 1$ and $x = 3$. Therefore, the area is given by the following.

$$\int_1^3 (e^x - (x - 1)) dx = \int_1^3 (e^x - x + 1) dx = \left(e^x - \frac{1}{2}x^2 + x \right) \Big|_1^3 = \left(e^3 - \frac{9}{2} + 3 \right) - \left(e - \frac{1}{2} + 1 \right) = e^3 - e - 2$$

29. What is the area of the region enclosed by the curve $y = \frac{2}{x}$, the x -axis, and the lines $x = 1$ and $x = 2$?

(A) $\ln 4 - \ln 2$

(B) $2 \ln 2$ ✓

(C) $\frac{3}{4}$

(D) $\frac{3}{2}$

Answer B

Correct. The graph of $y = \frac{2}{x}$ lies above the x -axis over the region in the first quadrant between the lines $x = 1$ and $x = 2$.

Therefore the area is given by $\int_1^2 \frac{2}{x} dx = 2 \ln x \Big|_1^2 = 2 \ln 2 - 2 \ln 1 = 2 \ln 2$.

8-4

30. What is the area of the region in the first quadrant that is bounded by the graph of $y = 2^x$, the horizontal line $y = 4$, and the y -axis?

(A) $8 - \frac{3}{\ln 2}$

(B) $\frac{3}{\ln 2}$

(C) $\frac{22}{3}$

(D) $\frac{15}{\ln 2}$

Answer A

Correct. The graph of $y = 2^x$ intersects the horizontal line $y = 4$ at $x = 2$. The top curve of the region is $y = 4$ and the bottom curve of the region is $y = 2^x$. Therefore the area is given by the following definite integral on the interval $[0, 2]$.

$$\int_0^2 (4 - 2^x) dx = \left(4x - \frac{2^x}{\ln 2} \right) \Big|_0^2 = \left(8 - \frac{4}{\ln 2} \right) - \left(0 - \frac{1}{\ln 2} \right) = 8 - \frac{3}{\ln 2}$$

31. $\int_{-1}^1 \frac{x^2 - x}{x} dx$ is

(A) -2

(B) 0

(C) $\frac{4}{3}$

(D) nonexistent

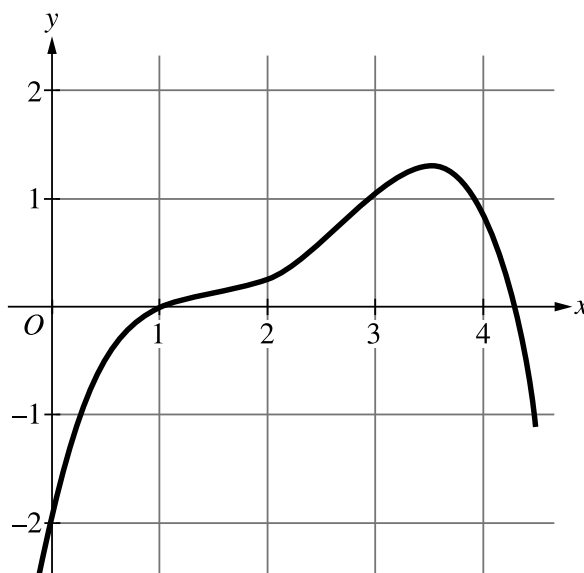
Answer A

Correct. The function $f(x) = \frac{x^2 - x}{x}$ has a removable discontinuity at $x = 0$, since $f(0)$ is undefined but $\lim_{x \rightarrow 0} \frac{x^2 - x}{x} = \lim_{x \rightarrow 0} (x - 1) = -1$. The definition of the definite integral can be extended to functions with removable discontinuities. If $g(x) = x - 1$, then $g(x) = f(x)$ for all x except $x = 0$, and therefore

$$\int_{-1}^1 f(x) dx = \int_{-1}^1 g(x) dx = \int_{-1}^1 (x - 1) dx = \left(\frac{1}{2}x^2 - x \right) \Big|_{-1}^1 = \left(\frac{1}{2} - 1 \right) - \left(\frac{1}{2} + 1 \right) = -2.$$

8-4

32.

Graph of f'

The graph of f' , the derivative of the function f , is shown. Which of the following differences is greatest?

- (A) $f(1) - f(0)$
(B) $f(2) - f(1)$
(C) $f(3) - f(2)$
(D) $f(4) - f(3)$

**Answer D**

Correct. The Fundamental Theorem of Calculus allows each difference to be expressed in terms of a definite integral, and the value of each definite integral can be estimated using the area of the region bounded by the graph of f' and the x -axis.

$$f(1) - f(0) = \int_0^1 f'(x) dx < 0, \text{ since the graph of } f' \text{ is below the } x\text{-axis on } [0, 1].$$

$$0 < f(2) - f(1) = \int_1^2 f'(x) dx < \frac{1}{2} \left(\frac{1}{2} \right) (1) = \frac{1}{4}$$

$$\frac{1}{2} < f(3) - f(2) = \int_2^3 f'(x) dx < 1$$

$$f(4) - f(3) = \int_3^4 f'(x) dx > 1$$

Therefore, the difference $f(4) - f(3)$ is greatest.